

Air Quality Health Index (AQHI) — Edmonton

2.0

Current AQHI at Edmonton Lendrum (18.14 km from venue)
Forecast Sun 9:00 AM: **2.1**

Health Risk	AQHI	At-Risk Population	General Population
Low Risk	2.1	Enjoy your usual outdoor activities.	Ideal air quality for outdoor activities.

No active Environment Canada weather alerts for the venue.

Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 15.0°C and feels near 14.5°C. Winds are 10 km/h from W, gusting near 13 km/h.
- The nearest current AQHI is 2 at Edmonton Lendrum, 18.1 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 3 (forecast tonight: 4, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High) puts the area near 3.
- The nearest air monitoring station (Leduc Sensor, 12.7 km) reports fine particulate matter (PM2.5) at 26.0 µg/m³.
- A nearby community sensor ('ACA_RMA', 7.4 km) reads 33.3 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 136.8 km to the SW, i.e. arriving from the SW.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-22) is 20.8 km N of the venue. The modeled back-trajectory does not pass near this fire (nearest approach 32.1 km), so this specific fire is unlikely to be the source of any smoke reaching the venue right now.
- The AQHI forecast for Sun 9:00 AM is 2.
- Thunderstorm conditions appear in the hourly forecast beginning around Sun 5:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are precipitation, thunderstorm.

Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

Hazard Snapshot — Overall Risk: High Risk

AIR QUALITY Low Risk 2.1 AQHI	HEAT Low Risk 26.2 °C apparent/humidex	WIND Low Risk 43.2 km/h peak gust	PRECIPITATION High Risk 1.6 mm/h maximum	THUNDERSTORM High Risk Sun 5:00 PM first forecast signal
AQHI RATE OF CHANGE Low Risk 0.0 AQHI change/hour			WIND SHEAR Low Risk 7.1 ° direction diff (10m vs 120m)	

Local Monitoring

Nearest station 2.0 Edmonton Lendrum · 18.14 km	Blended estimate 3.0 confidence High	PM2.5 (station) 26.0 µg/m³ Leduc Sensor · 12.73 km	PM2.5 (community) 33.3 µg/m³ ACA_RMA · 7.4 km
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